

erwise. Now, any analogue storage system can deteriorate under humid conditions. Also, maintenance of these tapes needs considerable time and money. If the hospital has a large number of records then indexing, labelling and retrieval of data would need a separate department. What do you do in such a situation—computerise?

However, if the data to be stored is huge, you would also need to compress it. As such, compression involves some loss of quality, the loss being directly proportional to the compression ratio. This is where the new technology comes into play. Called 'EchoView Convert', this compression technique was developed by the Singapore-based MatrixView Technologies who also have a research centre in Chennai.

MatrixView claims its compression technique as the "world's first completely digital echocardiogram storage and retrieval system at mathematically lossless quality and high compression ratios."

A Step Towards The Future

Sri Sathya Sai Institute of Higher Medical Sciences at Whitefield, on the outskirts of Bangalore, is one of MatrixView's clients in India. The institute is a hospital set up by the Sathya Sai Trust. Though the hospital renders its services for free, it has state-of-the-art equipment and technologies in place.

Keeping with the idea of providing the highest quality medical aid for free even to the poorest, the hospital converted all its VHS (analogue tapes), and patient records into a digitised format. This allows doctors to pull up patients' records, at any given time, on their systems. The hospital also has a 1 Gbps LAN (Local Area Network) connection.

To facilitate the conversion and archiving of analogue tapes is where MatrixView came into the picture with their proprietary EchoView system. Recommended by a visiting surgeon from the USA, who saw EchoView in action in a few Australian hospitals, Sri Sathya Sai Institute decided to do away with its analogue archiving systems in favour of the new technology.

As mentioned earlier, EchoView is a hardware solution that captures the Echocardiogram directly while the scan is in progress. These images are then optimised with ABO (Adaptive Binary Optimisation), a proprietary compression technology, and archived in MatrixView's .mvu format. Using an existing LAN network, such images are easily retrieved.

The Upside

Does EchoView help the institute to get the desired results? What about the cost involved? How does it benefit the patients and the hospital? Praveen Vemula, Technical Officer in charge of Hospital Administration at the institute answers, "EchoView has reduced costs and increased convenience. Doctors can now take an echocardiograph of every patient with a heart ailment and refer to it by just logging on to the local network. This ensures an easier tracking of a patient's record."

The system has been in use for six months now. The personnel of the Cardio OPD (out patients department) of the hospital, which sees



The compression ratio is so high that even after a high inflow of patients, storing all their echos and ultrasound scans is not a problem"

Praveen Vemula
Technical Officer,
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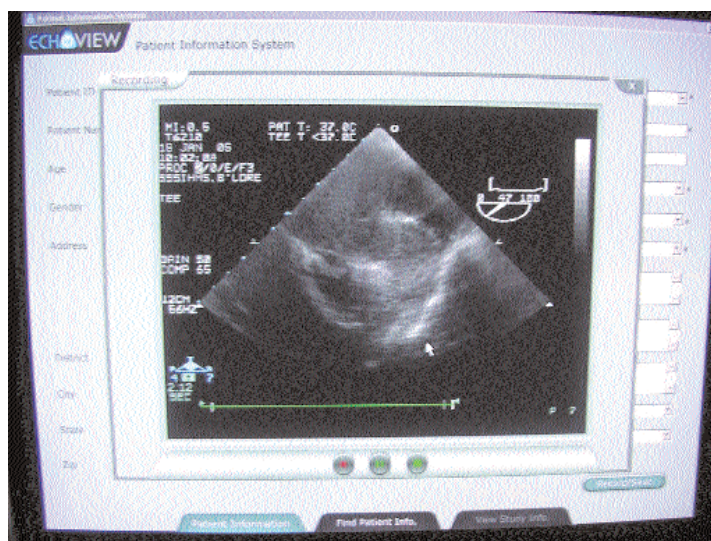
Compression Ratio Comparison			
	EchoView	JPG	JP2
Average Compression Ratio	32.9 times	7.3 times	8.5 times

Source: Ernst & Young Test Report

a large number of patients daily, are highly impressed. However, what about the USP of software—compression of all the data with almost zero loss in quality? "Earlier we used the JPEG compression technology. This afforded us a compression of up to five per cent the original file size," says Praveen.

Medical images and files such as an echocardiogram are highly detailed, and therefore, huge. Typically, each raw image, which means each file (an MRI scan or an echo), is around 2-3 GB. A five per cent compression still leaves a file size of around 1 GB. With EchoView, which uses the ABO compression technology, the image can be compressed by around 35 times.

So instead of storing a 3 GB file, you're left with a 70 MB file. "Considering the Cardio OPD sees close to 120 patients a day, the EchoView system has saved us a huge amount of disk space, without affecting the quality of the image," opines Praveen.



Doctors can view the patients medical reports in detail

The Flipside

Coupled with space saving, minimal loss of quality, zero maintenance cost and the 24-hour accessibility, EchoView is a killer application. What could the possible grouses be? "Doctors prefer the software to be customised as per their requirements. This is taking a bit longer. Being a turnkey application this problem was expected. The hardware compatibility issues that cropped up were due to limitations of the hardware than any glitch in the software," says Praveen.

With digitised archiving of important records such as the echo and ultrasound scans, Sri Sathya Institute has taken a step further in keeping its equipment and technologies among the best in the world. ■

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